

SECTION 1: System Architecture

Q1. Explain the complete architecture of your project.

Answer:

Our project follows a client-server architecture.

The workflow is:

User



React Frontend / Flutter App



Node.js + Express.js



FastAPI (ML Backend)



Machine Learning Models
(Random Forest)



MongoDB Database



External APIs
(OpenWeather, Government API,
Gemini API, Groq API)

The React frontend collects user input such as soil nutrients, pH, rainfall, and temperature.

This data is sent to the backend using REST APIs.

The backend processes the request. If prediction is required, it sends the data to FastAPI where the trained ML model predicts the result.

Finally, the prediction is returned to React and displayed to the user.

Q2. Why did you use two backends (Node.js and FastAPI)?

Answer:

Node.js is used for:

- User Authentication
- Database Operations
- CRUD APIs
- Community Posts
- Farm Ledger
- Admin Panel

FastAPI is used only for Machine Learning because

- Python supports ML libraries.
- TensorFlow and Scikit-Learn work efficiently in Python.
- FastAPI is very fast.
- Easy integration with ML models.

This separation makes the project modular.

Q3. Why not implement everything in Node.js?

Answer:

Machine Learning libraries such as

- Scikit-Learn
- TensorFlow
- NumPy
- Pandas

are designed mainly for Python.
Implementing ML in Node.js is difficult and less efficient.
Therefore we used FastAPI.

SECTION 2: Machine Learning

Q4. Why did you choose Random Forest?

Answer:

Random Forest has several advantages.

- High Accuracy
- Less Overfitting
- Handles large datasets
- Handles missing values better
- Works well with nonlinear data
- Gives stable predictions

Since agricultural data is complex, Random Forest performs better than a single Decision Tree.

Q5. Explain Random Forest.

Answer:

Random Forest is an Ensemble Learning Algorithm.

Instead of creating one decision tree, it creates many decision trees.

Each tree predicts the result independently.

Finally,

Regression

Average of all trees

Classification

Majority Voting

This improves accuracy.

Q6. Explain Random Forest mathematically.

Answer:

Suppose there are

100 Decision Trees.

Each tree predicts

Tree1 → Rice

Tree2 → Rice

Tree3 → Wheat

Tree4 → Rice

Majority Voting

Rice = 75 votes

Wheat = 25 votes

Final Prediction

Rice

For Yield Prediction

Tree1 = 38

Tree2 = 40

Tree3 = 42

Average

$(38+40+42) / 3$

=40

Q7. Why Random Forest instead of Decision Tree?

Answer:

Decision Tree

Only one tree

More overfitting

Less accurate

Random Forest

Many trees

Less overfitting

Higher accuracy

Better generalization

Q8. What is Ensemble Learning?

Answer:

Ensemble Learning means combining multiple models to produce one better prediction.

Random Forest is an example of Ensemble Learning.

Q9. What features are used for Crop Recommendation?

Answer:

The input features are

- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Soil pH
- Rainfall
- Temperature
- Humidity

The model predicts the most suitable crop.

Q10. What features are used for Yield Prediction?

Answer:

The model uses

- Soil Moisture
- Temperature
- Rainfall
- Humidity
- Soil pH
- Crop Duration
- NDVI (if available)

The output is predicted crop yield.

Q11. Why is Yield Prediction a Regression problem?**Answer:**

Because the output is a continuous numerical value like
4.8 tons/hectare

6.2 tons/hectare

Regression predicts numerical values.

Q12. Why is Fertilizer Recommendation Classification?**Answer:**

Because the output is one fertilizer category.

Example

Urea

DAP

NPK

Compost

These are categories.

Hence Classification.

SECTION 3: Model Training

Q13. Explain the ML workflow.**Answer:**

Dataset Collection

↓

Cleaning

↓

Feature Selection

↓

Preprocessing

↓

Train-Test Split

↓

Model Training

↓

Model Evaluation

↓

Save Model

↓

Deployment

↓

Prediction

Q14. What preprocessing did you perform?

Answer:

- Removed missing values
- Removed duplicate records
- Normalized numerical values
- Converted categorical values into numerical form where needed

Q15. What is Train-Test Split?

Answer:

The dataset is divided into two parts.

80%

Training

20%

Testing

Training teaches the model.

Testing checks performance on unseen data.

Q16. Why is Testing important?

Answer:

Testing tells whether the model performs well on new data.

Without testing we cannot judge the model's performance.

Q17. What evaluation metrics did you use?

Answer:

Crop Recommendation

Accuracy

Precision

Recall

F1 Score

Yield Prediction

RMSE

MAE

R² Score

Q18. What is Overfitting?

Answer:

The model memorizes training data and fails on new data.

Q19. How does Random Forest reduce Overfitting?

Answer:

It trains many different decision trees on different subsets of data and combines their predictions. Because the final result is based on all trees instead of one, errors from individual trees are reduced, leading to better generalization.

SECTION 4: FastAPI

Q20. Why FastAPI?

Answer:

FastAPI is

- Very Fast
 - Lightweight
 - Supports Async Programming
 - Automatically Generates Swagger Documentation
 - Easy integration with Python ML models
-

Q21. What happens when a user clicks Predict?

Answer:

1. React collects user inputs.
 2. Sends a POST request to Node.js.
 3. Node.js forwards prediction data to FastAPI.
 4. FastAPI loads the trained ML model.
 5. Model predicts the result.
 6. FastAPI returns JSON.
 7. React displays the prediction.
-

Q22. Why save the model as a file?

Answer:

Training every time is very slow.

So we train once and save the model using `.pkl` or `.keras`.

During prediction we simply load the saved model.

SECTION 5: APIs

Q23. Which APIs are used?

Answer:

- OpenWeather API
 - Google Gemini API
 - Groq API
 - Government Open Data API
-

Q24. Why OpenWeather API?

Answer:

It provides

- Temperature
- Humidity
- Rainfall

- Wind Speed
- Weather Conditions

These values help in crop recommendation and season planning.

Q25. Why Government Open Data API?

Answer:

It provides real-time agricultural information such as market prices, crop statistics, and government datasets, making the recommendations more reliable and up to date.

Q26. Why Gemini API?

Answer:

Gemini provides AI-based responses for user queries. Farmers can ask questions in natural language about crops, diseases, fertilizers, or farming practices and receive detailed guidance.

Q27. Why Groq API?

Answer:

Groq is used as an AI inference service to generate responses quickly with low latency. It improves chatbot response speed and can serve as an alternative or backup AI service.

Q28. What is a REST API?

Answer:

A REST API is a way for different applications to communicate over HTTP using methods like GET, POST, PUT, and DELETE. In our project, the frontend communicates with the backend using REST APIs.

Q29. Why use JSON?

Answer:

JSON is lightweight, easy to read, language-independent, and is the standard format for exchanging data between the frontend and backend.

SECTION 6: Database

Q30. Why MongoDB instead of MySQL?

Answer:

MongoDB stores data as JSON-like documents, making it flexible for different data structures. Since our application has users, posts, chat history, ledger entries, and other varying data, MongoDB is more suitable than a relational database.

Q1. What is Machine Learning?

Answer:

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed. Instead of writing rules for every situation, we train the model using historical data so it can recognize patterns and predict future outcomes.

In our project, Machine Learning is used for:

- Crop Recommendation
 - Yield Prediction
 - Fertilizer Recommendation
-

Q2. Why did you use Machine Learning in your project?

Answer:

Agricultural data depends on many factors such as soil nutrients, pH, rainfall, humidity, and temperature. It is difficult to create fixed rules for every situation.

Machine Learning learns from previous agricultural data and predicts the best output for new input data, making recommendations more accurate.

Q3. What Machine Learning algorithms did you use?

Answer:

We used:

- Random Forest Classifier – for Crop Recommendation
- Random Forest Regressor – for Yield Prediction
- Random Forest Classifier – for Fertilizer Recommendation

Random Forest provides high accuracy and reduces overfitting.

Q4. Why did you choose Random Forest?

Answer:

We selected Random Forest because it:

- Gives high accuracy
- Reduces overfitting
- Works well with large datasets
- Handles nonlinear relationships
- Can process many input features
- Produces stable predictions

It performs better than a single Decision Tree for agricultural data.

Q5. What is Random Forest?

Answer:

Random Forest is an Ensemble Machine Learning algorithm.

Instead of using one Decision Tree, it creates many Decision Trees.

Each tree makes its own prediction.

The final result is obtained by:

- Majority Voting (Classification)
- Average Prediction (Regression)

This improves accuracy and reliability.

Q6. What is Ensemble Learning?

Answer:

Ensemble Learning is a technique where multiple Machine Learning models work together to make a better prediction.

Random Forest is an example of Ensemble Learning because it combines many Decision Trees.

Q7. Explain how Random Forest works.

Answer:

The working process is:

1. Collect training data.
2. Create multiple random subsets of the dataset.
3. Build one Decision Tree for each subset.
4. Each tree predicts the output independently.
5. Combine all predictions.
6. Return the final result.

This reduces errors and improves prediction accuracy.

Q8. What is a Decision Tree?

Answer:

A Decision Tree is a tree-like Machine Learning algorithm that makes decisions using conditions.

Each internal node represents a condition.

Each branch represents the result of that condition.

The leaf node gives the final prediction.

Q9. Difference between Decision Tree and Random Forest?

Answer:

Decision Tree	Random Forest
Single tree	Multiple trees
Faster training	Slightly slower
Can overfit	Less overfitting
Lower accuracy	Higher accuracy
Less reliable	More reliable

Q10. What is Classification?

Answer:

Classification predicts categories or classes.

Example:

Crop:

- Rice
- Wheat
- Cotton

Fertilizer:

- Urea
- DAP
- NPK

These outputs belong to fixed classes.

Q11. What is Regression?

Answer:

Regression predicts continuous numerical values.

Example:

Crop Yield

- 3.8 tons/hectare

- 5.4 tons/hectare

Since the output is numeric, regression is used.

Q12. Which modules use Classification?

Answer:

Classification is used in:

- Crop Recommendation
- Fertilizer Recommendation

Because the outputs are categories.

Q13. Which module uses Regression?

Answer:

Yield Prediction uses Regression because the output is a numerical value.

Q14. What are Features?

Answer:

Features are the input variables given to the model.

Examples:

- Nitrogen
- Phosphorus
- Potassium
- Soil pH
- Temperature
- Rainfall
- Humidity

These features help the model make predictions.

Q15. What is the Target Variable?

Answer:

The Target Variable is the output that the model predicts.

Examples:

Crop Recommendation → Crop Name

Yield Prediction → Crop Yield

Fertilizer Recommendation → Fertilizer Name

Q16. What dataset did you use?

Answer:

We used:

- Kaggle agricultural datasets
- Government agricultural datasets
- Soil data
- Weather data
- Crop production data

These datasets were cleaned before training.

Q17. Why is dataset quality important?

Answer:

Machine Learning models learn from data.

If the dataset contains errors, duplicates, or missing values, the model gives inaccurate predictions.

Good quality data improves accuracy.

Q18. What preprocessing did you perform?

Answer:

We performed:

- Missing value handling
 - Duplicate removal
 - Data cleaning
 - Feature selection
 - Normalization (where required)
-

Q19. What are Missing Values?

Answer:

Missing values are empty or unavailable values in the dataset.

Example:

Temperature = NULL

These values must be handled before training.

Q20. How did you handle Missing Values?

Answer:

Depending on the dataset, missing values can be:

- Removed
- Replaced with mean
- Replaced with median
- Replaced with mode

This ensures the dataset is complete.

Q21. What are Duplicate Records?

Answer:

Duplicate records are repeated rows in the dataset.

They increase bias and reduce model performance.

We removed duplicate entries.

Q22. What is Feature Selection?

Answer:

Feature Selection means choosing only the important input variables for prediction.

This reduces unnecessary data and improves model performance.

Q23. Why is Feature Selection important?

Answer:

It:

- Reduces training time

- Improves accuracy
 - Removes irrelevant features
 - Prevents overfitting
-

Q24. What is Training?

Answer:

Training is the process of teaching the Machine Learning model using historical data.

The model learns patterns and relationships between inputs and outputs.

Q25. What is Testing?

Answer:

Testing checks how well the trained model performs on new, unseen data.

It measures the model's performance.

Q26. What is Train-Test Split?

Answer:

The dataset is divided into:

- Training Data (usually 80%)
- Testing Data (usually 20%)

Training is used to build the model.

Testing is used to evaluate it.

Q27. Why not use the entire dataset for training?

Answer:

If we use all data for training, we cannot evaluate the model on unseen data.

Testing data is necessary to measure real-world performance.

Q28. What is Model Accuracy?

Answer:

Accuracy is the percentage of correct predictions made by the model.

Higher accuracy means better performance.

Q29. What is Overfitting?

Answer:

Overfitting happens when the model memorizes the training data instead of learning patterns.

It performs well on training data but poorly on new data.

Q30. What is Underfitting?

Answer:

Underfitting occurs when the model cannot learn enough from the training data.

As a result, it performs poorly on both training and testing data.

Q31. How does Random Forest reduce Overfitting?

Answer:

Random Forest builds many Decision Trees using different subsets of the data.

The final prediction is based on all trees instead of one.

This reduces the effect of individual errors and improves generalization.

Q32. What is Bootstrapping?

Answer:

Bootstrapping is a sampling technique where multiple random datasets are created from the original dataset with replacement.

Each Decision Tree is trained on a different bootstrap sample.

Q33. What is Majority Voting?

Answer:

In classification, each Decision Tree votes for a class.

The class with the highest number of votes becomes the final prediction.

Example:

Rice $\rightarrow$ 80 votes

Wheat $\rightarrow$ 15 votes

Cotton $\rightarrow$ 5 votes

Final Prediction = Rice

Q34. What is Averaging in Random Forest?

Answer:

For regression, each Decision Tree predicts a numerical value.

The final output is the average of all predictions.

Example:

Tree1 = 40

Tree2 = 42

Tree3 = 38

Final Prediction = $(40 + 42 + 38) / 3 = 40$

Q35. What evaluation metrics are used for Classification?

Answer:

Common metrics are:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

These help evaluate how well the classification model performs.

Q36. What evaluation metrics are used for Regression?

Answer:

Common metrics are:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R² Score (Coefficient of Determination)

These measure the difference between predicted and actual values.

Q37. Why didn't you use Neural Networks?

Answer:

Neural Networks require larger datasets, more computation, and longer training time.

Random Forest provided good accuracy with simpler implementation for our agricultural dataset.

Q38. Can Random Forest handle nonlinear data?

Answer:

Yes.

Random Forest can learn complex nonlinear relationships between features and outputs, making it suitable for agricultural prediction problems.

Q39. What are the limitations of Random Forest?

Answer:

Some limitations are:

- Requires more memory
- Training takes longer than a single Decision Tree
- Model size is larger
- Less interpretable than a Decision Tree

Despite these, it offers better accuracy and robustness.

Q40. Why is Random Forest suitable for AgriVista?

Answer:

Random Forest is suitable because agricultural data depends on many interacting factors such as soil nutrients, weather, rainfall, and temperature. It handles multiple features effectively, reduces overfitting, provides reliable predictions, and works well for both classification and regression tasks. This makes it an excellent choice for crop recommendation, fertilizer recommendation, and yield prediction in AgriVista.

Part 4: APIs, Database, Deployment & Security (Questions 81–100)

BE Computer Engineering Final Year External Viva (AgriVista Project)

APIs (Q81–Q88)

Q81. What is an API?

Answer:

API (Application Programming Interface) is a set of rules that allows two different software applications to communicate with each other.

In our project, the frontend communicates with the backend through APIs, and the backend also communicates with external services like OpenWeather and Gemini APIs.

Example:

The React frontend sends soil data to the backend API, and the backend returns the crop prediction.

Q82. What is a REST API?

Answer:

REST (Representational State Transfer) API is a web service that allows communication between client and server using HTTP methods.

Common HTTP Methods:

- GET – Retrieve data
- POST – Send data
- PUT – Update data
- DELETE – Delete data

REST APIs are lightweight, fast, and widely used in web applications.

Q83. Which APIs did you use in AgriVista?

Answer:

We used the following APIs:

- OpenWeather API – To fetch live weather information.
- Government Open Data API – To fetch real-time crop market prices.
- Google Gemini API – To provide AI-based farming guidance.
- Groq API – To generate fast AI responses for the chatbot.

These APIs make the system more dynamic by providing real-time information.

Q84. Why did you use the OpenWeather API?

Answer:

The OpenWeather API provides real-time weather information such as:

- Temperature
- Humidity
- Rainfall
- Wind Speed
- Weather Conditions

These parameters are important for crop recommendation and season planning.

Q85. Why did you use the Gemini API?

Answer:

Gemini API is used as an AI assistant.

Farmers can ask questions like:

- Which crop should I grow?
- How do I control plant diseases?
- Which fertilizer should I use?

Gemini understands natural language and provides intelligent responses.

Q86. Why did you use the Government Open Data API?

Answer:

Government Open Data API provides authentic agricultural information like:

- Market prices
- Crop statistics
- Agricultural datasets

Using official government data increases the reliability of the system.

Q87. Why did you use the Groq API?

Answer:

Groq API is used for fast AI inference.

It generates chatbot responses with lower latency and improves response speed compared to many traditional AI services.

Q88. Why is JSON used in APIs?

Answer:

JSON (JavaScript Object Notation) is used because:

- It is lightweight.
- Easy to read.
- Easy to parse.
- Language-independent.
- Fast for data transfer.

Example:

```
{
  "crop": "Rice",
  "yield": "4.8 Tons/Hectare"
}
```

Database (Q89–Q94)

Q89. Why did you choose MongoDB?

Answer:

MongoDB is a NoSQL database that stores data in JSON-like documents.

We selected MongoDB because:

- Flexible schema
 - Easy to store nested data
 - Faster development
 - Good support with Node.js
 - Scalable
-

Q90. What data is stored in MongoDB?

Answer:

MongoDB stores:

- User Information

- Login Credentials (hashed passwords)
- Farmer Posts
- Crop History
- Farm Ledger
- AI Chat History
- Notifications
- Season Planner Data
- Market Data (if cached)

Machine Learning models are stored separately, not inside MongoDB.

Q91. Why not use MySQL?

Answer:

MySQL is a relational database with a fixed schema.

Our application contains different types of data such as:

- Posts
- Chat messages
- Ledger
- Images
- User profiles

MongoDB handles this flexible data structure more efficiently.

Q92. What is a NoSQL Database?

Answer:

A NoSQL database stores data in flexible formats like documents instead of tables.

MongoDB stores data as JSON documents.

Example:

```
{
  "name": "Rahul",
  "village": "Ahmednagar",
  "crop": "Cotton"
}
```

Q93. How does React communicate with MongoDB?

Answer:

React never communicates directly with MongoDB.

Communication flow:

React

↓

Express.js

↓

Node.js

↓

MongoDB

Node.js receives requests from React and performs database operations using Mongoose.

Q94. What is Mongoose?**Answer:**

Mongoose is an ODM (Object Data Modeling) library for MongoDB.

It helps to:

- Create schemas
- Validate data
- Perform CRUD operations
- Connect Node.js with MongoDB

Deployment (Q95–Q98)

Q95. How can this project be deployed?**Answer:**

The project can be deployed using:

Frontend:

- Vercel
- Netlify

Backend:

- Render
- Railway
- AWS EC2

Database:

- MongoDB Atlas

Machine Learning API:

- Render
- Railway
- AWS

Mobile App:

- Google Play Store

Q96. What is Cloud Deployment?**Answer:**

Cloud deployment means hosting the application on cloud servers so users can access it from anywhere using the internet.

Instead of running on a local computer, the application runs on remote servers.

Q97. What is MongoDB Atlas?**Answer:**

MongoDB Atlas is the cloud version of MongoDB.

Advantages:

- Automatic backups
- Cloud hosting
- High availability
- Secure access
- Easy scalability

Q98. Why separate the ML Backend from the Node.js Backend?**Answer:**

We separated them because:

Node.js is good for:

- Authentication
- Database
- CRUD APIs

FastAPI is good for:

- Machine Learning
- TensorFlow
- Scikit-Learn
- Faster prediction

This modular architecture improves scalability, maintainability, and performance.

Security (Q99–Q100)

Q99. How did you secure your project?

Answer:

We implemented several security measures:

- JWT Authentication for secure login.
- Password hashing using bcrypt.
- Environment variables (.env) to hide API keys and database URLs.
- Input validation to prevent invalid data.
- CORS configuration to allow only trusted frontend domains.
- HTTPS support during deployment.
- Secure API communication between frontend and backend.

These measures help protect user data and reduce common web security risks.

Q100. If the external examiner asks, "How can you improve the security further?", what will you answer?

Answer:

Future security improvements include:

- Two-Factor Authentication (2FA)
- Role-Based Access Control (Admin/Farmer)
- Rate limiting to prevent API abuse
- CAPTCHA to stop bots
- Database encryption for sensitive information
- Regular security audits
- Secure logging and monitoring
- Automated backups and disaster recovery
- OAuth login (Google or Microsoft)
- Web Application Firewall (WAF)

These enhancements would make the system more secure and suitable for production use.